

Project Report: Support Dashboard Analysis (Uke 24)

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Abstract

This report presents an analysis of customer support interactions for MORSE, a telecommunications company, based on data collected during Week 24. The objective was to develop a support dashboard that provides insights into daily inquiry volumes, call durations, and customer satisfaction. Key findings include the distribution of inquiries across weekdays and time blocks, average and extreme call durations, and the calculation of the Net Promoter Score (NPS). The analysis highlights operational patterns and customer sentiment, offering actionable insights for service improvement.

1. Introduction

Effective customer support is paramount for maintaining customer satisfaction and operational efficiency in the telecommunications sector. This project addresses the need for a comprehensive support dashboard for MORSE, focusing on data from Week 24. The dashboard aims to provide a clear overview of support activities, enabling informed decision-making regarding resource allocation and service quality enhancements. The methodology involves data extraction, statistical analysis, and visualization, culminating in a Net Promoter Score assessment to gauge customer loyalty.

2. Methodology

Data for this project was sourced from `support_uke_24.xlsx`, a Microsoft Excel spreadsheet containing records of customer support interactions during Week 24. The dataset included information on the day of the week (`ukedag`), time of call (`klokkeslett`), call duration (`varighet`), and customer satisfaction scores (`tilfredshet`).

The analysis was conducted using Python, leveraging the `pandas` library for data manipulation, `matplotlib` for data visualization, and `numpy` for numerical operations. The process involved several distinct steps:

- 1. Data Loading:** The Excel file was read into a pandas DataFrame, and relevant columns were extracted into separate arrays for processing.
- 2. Inquiry Distribution Analysis:** The number of inquiries for each weekday was calculated and visualized using a bar chart to identify peak service days.
- 3. Call Duration Analysis:** Minimum, maximum, and average call durations were computed to understand the range and central tendency of service times.
- 4. Time Block Analysis:** Inquiries were categorized into two-hour blocks (08-10, 10-12, 12-14, 14-16) and visualized using a pie chart to pinpoint busy periods within the day.
- 5. Net Promoter Score (NPS) Calculation:** Customer satisfaction scores were used to calculate the NPS, classifying customers as Promoters (score 9-10), Passives (score 7-8), or Detractors (score 1-6). Scores without feedback were excluded from this calculation.

All visualizations were saved as PNG image files, and the numerical results were presented in a clear, concise manner.

3. Results

3.1. Inquiry Distribution by Weekday

The analysis of inquiry distribution across Week 24 revealed varying call volumes throughout the week. Monday recorded the highest number of inquiries, followed by

Tuesday and Wednesday, indicating a typical start-of-week surge in customer contact. Friday and Thursday had comparatively lower volumes.

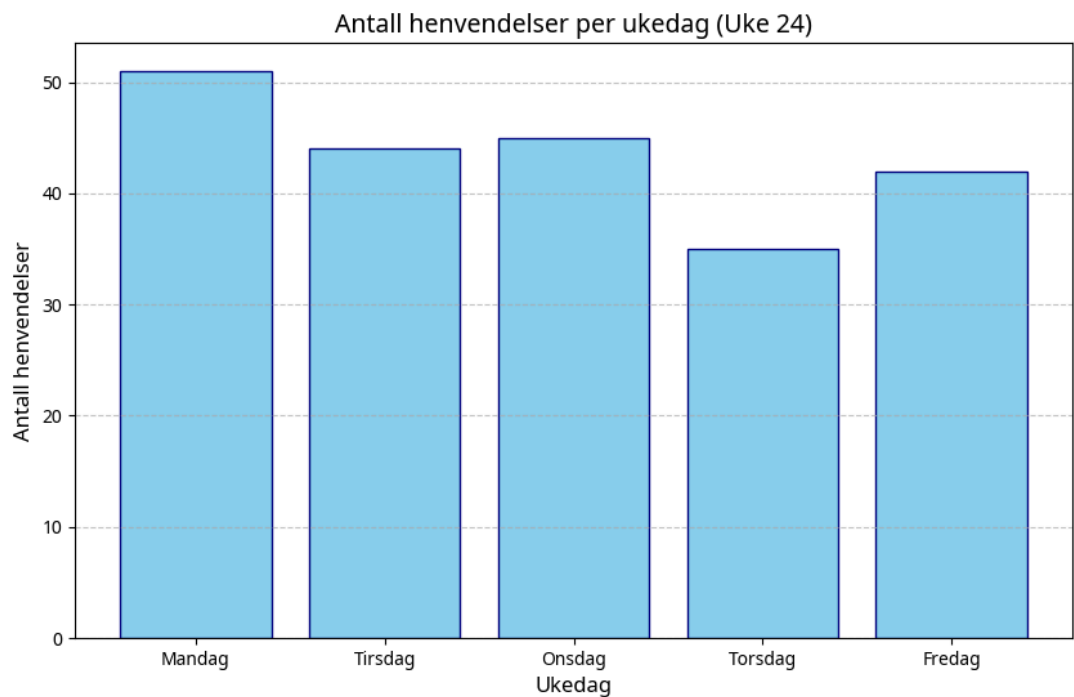


Figure 1. Distribution of customer inquiries across weekdays in Week 24.

3.2. Call Duration Statistics

The call duration analysis provided insights into the length of customer interactions. The shortest recorded call was 00:00:59, while the longest extended to 00:11:28. The average call duration was calculated to be approximately 00:06:40. These metrics are crucial for assessing agent efficiency and identifying potential areas for process optimization.

3.3. Inquiry Distribution by Time Block

Customer inquiries were further analyzed by two-hour time blocks to identify peak operational hours. The period between 08:00 and 10:00 experienced the highest volume of calls, accounting for 45.2% of all inquiries. This was followed by the 12:00-14:00 block (20.7%), 10:00-12:00 block (18.4%), and 14:00-16:00 block (15.7%). This distribution suggests that early morning hours are the busiest for the support team.

Henvendelser fordelt på tidsrom (Uke 24)

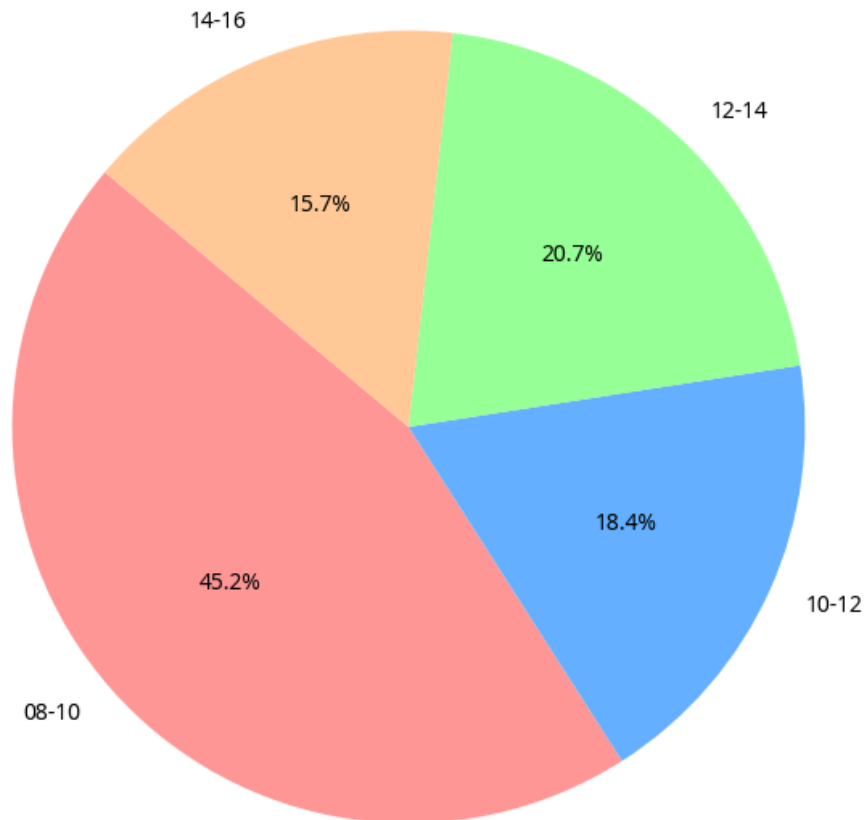


Figure 2. Percentage distribution of customer inquiries across two-hour time blocks in Week 24.

3.4. Net Promoter Score (NPS)

The Net Promoter Score (NPS) was calculated based on customer satisfaction ratings. Out of 43 valid responses, 65.1% were classified as Promoters, 11.6% as Passives, and 23.3% as Detractors. This resulted in a calculated NPS of 41.9. An NPS of 41.9 indicates a strong positive sentiment among customers, suggesting a good level of customer loyalty and satisfaction, though there is still room for improvement by addressing the concerns of Detractors.

4. Discussion

The findings from this support dashboard analysis offer valuable insights into MORSE's customer service operations. The concentration of inquiries on Mondays and during early morning hours (08:00-10:00) suggests a need for optimized staffing schedules to match demand. The average call duration provides a benchmark for agent performance, while the range highlights the variability in customer issues. The positive NPS score is encouraging, reflecting a generally satisfied customer base. However, the presence of Detractors indicates specific areas where service quality or product experience could be enhanced to convert these customers into Promoters. Further investigation into the nature of calls from Detractors could yield targeted improvement strategies.

5. Conclusion

This project successfully developed a support dashboard providing a comprehensive overview of MORSE's customer support activities for Week 24. The analysis of inquiry patterns, call durations, and customer satisfaction through NPS offers a robust foundation for operational improvements. By leveraging these insights, MORSE can strategically allocate resources, refine service processes, and ultimately enhance customer experience and loyalty.

References

- [1] Universitetet i Sørøst-Norge. (n.d.). *Prosjektoppgave i PY1010*. [Project Assignment Document]. [2] Blueprnt. (2018, September 17). *Net Promoter Score*. Retrieved from <https://www.blueprnt.com/2018/09/17/net-promoter-score/>